

OPD Normalization Strategies: Pseudocode

Shared Setup

For each rollout chunk index t , rollout sample b , and flattened action token k , OPD first computes the sampled-action teacher margin

$$\Delta_{t,b,k} = \log \pi_{\text{teacher}}(a_{t,b,k} \mid s_{t,b}) - \log \pi_{\text{student,old}}(a_{t,b,k} \mid s_{t,b}).$$

Let $M_{t,b,k} \in \{0, 1\}$ be the valid-token mask. In all modes below, invalid tokens are zeroed at the end:

$$A \leftarrow A \odot M.$$

For OpenVLA-OFT chunked actions, the last dimension is typically

$$k = (h, d), \quad h \in \{1, \dots, H\}, \quad d \in \{1, \dots, D\},$$

where $H = 8$ future actions and $D = 7$ action dimensions.

Mode Summary

Mode	What Is Normalized	Dense Token Feedback Preserved?
<code>group_zscore</code>	chunk scalar within rollout group	No
<code>token_zscore</code>	each flattened token position k	Yes
<code>action_dim_zscore</code>	each action dimension d	Yes
<code>positive_clip</code>	raw positive token margins	Yes
<code>teacher_prob</code>	sampled teacher probability weights	Yes

1. Current `group_zscore`

Difference: collapse all action tokens in a chunk into one scalar, normalize that scalar within a rollout group, then broadcast it back to every token.

1. Compute one score per chunk/sample by averaging token margins:

$$s_{t,b} = \frac{\sum_k M_{t,b,k} \Delta_{t,b,k}}{\max(1, \sum_k M_{t,b,k})}.$$

2. Partition the batch dimension into groups g of size G :

$$b = (g, i), \quad i \in \{1, \dots, G\}.$$

3. For every chunk t and group g , compute:

$$\mu_{t,g} = \frac{1}{G} \sum_{i=1}^G s_{t,g,i}, \quad \sigma_{t,g} = \text{std}_{i=1}^G(s_{t,g,i}).$$

4. Broadcast the normalized scalar to every token in that chunk/sample:

$$A_{t,g,i,k} = \frac{s_{t,g,i} - \mu_{t,g}}{\sigma_{t,g} + \epsilon}.$$

In one line:

token margins $\rightarrow$ chunk mean $\rightarrow$ group z-score $\rightarrow$ same advantage for all tokens in the chunk.

2. token_zscore

Difference: keep each flattened token position separate. Do not average the action chunk into one scalar.

For each token position k :

$$\mu_k = \frac{\sum_{t,b} M_{t,b,k} \Delta_{t,b,k}}{\max\left(1, \sum_{t,b} M_{t,b,k}\right)},$$
$$\sigma_k = \sqrt{\frac{\sum_{t,b} M_{t,b,k} (\Delta_{t,b,k} - \mu_k)^2}{\max\left(1, \sum_{t,b} M_{t,b,k}\right)}}.$$

Then:

$$A_{t,b,k} = \frac{\Delta_{t,b,k} - \mu_k}{\sigma_k + \epsilon}.$$

This preserves token-level credit assignment:

token k_1 and token k_2 may receive different OPD feedback inside the same chunk.

3. action_dim_zscore

Difference: keep semantic action dimensions separate. Pool over chunk horizon h , rollout chunks t , and batch samples b , but not over action dimension d .

Reshape:

$$\Delta_{t,b,k} \rightarrow \Delta_{t,b,h,d}.$$

For each action dimension d :

$$\mu_d = \frac{\sum_{t,b,h} M_{t,b,h,d} \Delta_{t,b,h,d}}{\max\left(1, \sum_{t,b,h} M_{t,b,h,d}\right)},$$
$$\sigma_d = \sqrt{\frac{\sum_{t,b,h} M_{t,b,h,d} (\Delta_{t,b,h,d} - \mu_d)^2}{\max\left(1, \sum_{t,b,h} M_{t,b,h,d}\right)}}.$$

Then:

$$A_{t,b,h,d} = \frac{\Delta_{t,b,h,d} - \mu_d}{\sigma_d + \epsilon}.$$

This avoids mixing, for example, gripper margins with translation or rotation margins.

4. positive_clip

Difference: no centering and no z-score. Preserve raw token-level margins, but only reinforce actions that the teacher scores above the old student.

For each token:

$$A_{t,b,k} = \min(c, \max(0, \Delta_{t,b,k})).$$

Interpretation:

$$\Delta_{t,b,k} > 0 \Rightarrow \text{reinforce sampled action token,}$$

$$\Delta_{t,b,k} \leq 0 \Rightarrow \text{do not actively push against it.}$$

This is conservative sampled imitation and may be useful when the teacher is strong on the current task but unreliable for retained old-task behavior.

5. teacher_prob

Difference: ignore the student margin and weight sampled actions by teacher confidence. This is not full-distribution KL; it is sampled-action distillation.

For each sampled action token:

$$w_{t,b,k} = \exp(\text{clip}(\log \pi_{\text{teacher}}(a_{t,b,k} \mid s_{t,b}), -20, 0)).$$

Normalize the scale over valid tokens:

$$\bar{w} = \frac{\sum_{t,b,k} M_{t,b,k} w_{t,b,k}}{\max\left(1, \sum_{t,b,k} M_{t,b,k}\right)}.$$

Then:

$$A_{t,b,k} = \frac{w_{t,b,k}}{\bar{w} + \epsilon}.$$

Interpretation:

high teacher probability $\Rightarrow$ larger imitation weight; low teacher probability $\Rightarrow$ small update.

Continual-Learning Reading

With a current-task SFT teacher, `group_zscore`, `token_zscore`, and `action_dim_zscore` are still relative/centered objectives: some sampled actions receive negative advantages even if the teacher is generally good. In contrast, `positive_clip` and `teacher_prob` are more conservative imitation signals: they mostly reinforce what the current-task teacher likes, but they do not by themselves provide old-task retention signal. Forgetting prevention still requires replay, old-task teacher/base regularization, EWC, or another source of old-task pressure.